Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

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SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	K&Z
Project / Solution Name	Domain 1
Name (Leader)	Chong Wen Kai
Email & Phone (Leader)	chongwenkai0@gmail.com 0123909932
University Affiliation(s)	Xiamen University Malaysia IMU (International Medical University)
Live Prototype URL	https://kn-z-2.replit.app/
Code Repository URL (public / access granted to judges)	https://github.com/Kai9932/KnZ-SDC https://github.com/Kai9932/KnZ-Runpod

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

K&Z connects the user interface, backend services, AI models, external AI APIs and labour market datasets to deliver personalized career guidance and future-skills recommendation

Users interact with the web-based interface to complete the RIASEC career assessment and upload a resume for analysis. The frontend sends user inputs to the python FastAPI backend, which coordinates the relevant processing functions. The resume analysis module uses a pre-trained DistilBERT model to classify resume information, while the career-matching module compares user' skills interest and RIASEC profiles with relevant occupations

The system integrates public labour-market dataset like DOSM and O*NET occupation information support career recommendation, salary or skills-related insights and RIASEC dataset. The DeepSeek API provides AI-generated explanations and contextual analysis to help users understand career options, emerging skills, and potential skills gaps. Based on these results, K&Z present career matches and personalized reskilling recommendation through the frontend.

The application is developed using python and FastAPI with supporting machine learning libraries and pre-trained models. Replit is used for development workflows while RunPod can host computationally demanding models or inference services where required.

Overall the system combines user input, AI-based analysis, public labor-market evidence, and generative AI explanations to provide an integrated, data-informed career guidance experience.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python	Main programming language for backend development, AI processing and career matching logic	Python Software Foundation (PSF) License

Name & Version	Purpose in Project	License
HTML	Structures the web interface and career guidance pages	WHATWG HTML Standard; not a third-party software library
CSS	Styles the website and user interface	W3C CSS specifications; not a third-party software library
JavaScript	Implements frontend interactions and communicates with backend endpoints	ECMAScript specification; not a third-party software library
FastAPI	Open-source python framework for building backend APIs and serving application routes	MIT license
PyTorch	Machine-learning framework used to run supported AI models	BSD-style License (main project)
Hugging Face transformer	Library for loading and using pre-trained transformer models including DistilBERT	Apache License 2.0
Scikit-learn	Machine-learning library used for data processing and relevant classification or matching function	BSD 3-Clause License
Numpy	Support numerical operation and data processing	BSD 3-Clause License
Joblib	Saves and loads supported trained machine-learning models and related objects	BSD 3-Clause License
DeepseekAPI	External AI service used to generate contextual explanation and career-related insights	Subject to DeepSeek's applicable API terms of service

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Deepseek API	Generates contextual AI explanations and career-related insights, including support for interpreting career options, skills gaps, and future-skills information.	Paid
RunPod API — RunPod	Used only if the application calls RunPod's API to submit inference requests to a hosted AI model.	Sponsor-provided credits
GoogleAuth API	Google login	Paid

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit	Development environment and intended web application deployment for the FastAP	Hackathon-sponsored credits
Runpod	Intended hosting and inference for computationally demanding AI models, keeping large model files separate from the main GitHub repository.	Hackathon-sponsored credits

Service & Provider	Purpose in Project	Plan / Region
Github	Source-code version control and repository hosting. Large model files and static assets may be stored separately where necessary.	Pro (self-paid)

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
DistilBERT	HuggingFace	fine-tuned	Apache License 2.0
DeepSeek model	Deepseek	Access through external API	Subject to Deepseek's applicable API terms
RIASEC career matching logic	Developed by K&Z	Self-trained	Team developed code; but third party dataset
Resume classification	Developed by K&Z	Self-trained	Team developed code; but third party dataset

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Resume Dataset PDF	https://www.kaggle.com/datasets/hadikp/resume-data-pdf	Public	MIT	n

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Annual Salaries & Wages by Industry & Sex	Annual Salaries & Wages by Industry & Sex | OpenDOSM	Public	Creative Commons Attribution 4.0 International License (CC BY 4.0)	No
Annual Salaries & Wages by Skill Category & Sex	Annual Salaries & Wages by Skill Category & Sex | OpenDOSM	Public	Creative Commons Attribution 4.0 International License (CC BY 4.0)	No
Annual Salaries & Wages by Occupation & Sex	Annual Salaries & Wages by Occupation & Sex | OpenDOSM	Public	Creative Commons Attribution 4.0 International License (CC BY 4.0)	no
O*NET® 31.0 Database (all dataset from there)	https://www.onetcenter.org/database.html	Public	Creative Commons Attribution 4.0 International License	No

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

✓ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Gemini AI	Image generation and video generation	Moderate
Chatgpt	Debugging and design of UI	Moderate
Claude AI	Backend knowledge	Minor

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Our team used Gemini AI to generate images and videos, ChatGPT to debug code and improve UI styling and visual effects, and Claude AI to explore smarter development approaches. We reviewed AI-generated suggestions, checked code by running and testing the application, and refined outputs to fit our project requirements. We did not blindly accept AI responses; team members evaluated their relevance, correctness, and usability before implementation. Our team remains responsible for the final code, design decisions, functionality, and project outcomes.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

✓ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Credits	Deployment of software
Runpod	credits	Storing models

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

It is essential for our project as with github premium account the models are too big to be stored, with runpod it helped to stored the models. While Replit allows our group to deploy the app eventhough the device is shutdown, unlike uvicorn

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

K&Z may process user-provided resumes and RIASEC questionnaire responses to generate personalised career recommendations, skills-gap analysis, and reskilling suggestions. Resume content may be processed by the backend and AI models, while relevant prompts or information may be sent to the DeepSeek API for AI-generated explanations. Public labour-market datasets are also used. Users should avoid submitting sensitive personal information during demonstrations. Access to API credentials is restricted through environment variables, and secrets are not included in public source code. Any assessment results stored by the application should be protected against unauthorised access and retained only as long as necessary. The team will use synthetic or sample profiles for hackathon demonstrations rather than real personal resumes. Data deletion and retention procedures will depend on the storage and deployment configuration.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

K&Z depends on the accuracy and coverage of public labour-market datasets, which may not fully reflect rapidly changing job requirements. DeepSeek-generated explanations may contain inaccurate information or unsupported claims, while resume classification and career matching may be affected by model bias or incomplete user information. AI recommendations are intended as decision-support guidance, not guarantees of employment, salary, or future job demand. The system combines model outputs with available occupational and skills data, and the team reviews generated content and tests application functionality before implementation. Recommendations should be interpreted cautiously and verified against reliable sources. The application also depends on external API availability, usage limits, and service terms. Further validation with diverse user profiles and updated labour-market data is needed to improve reliability and fairness.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ✓ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ✓ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ✓ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ✓ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ✓ Any use of official SDC sponsor resources has been declared in Section 5.
- ✓ The prototype described here is accessible to judges through the links provided in Section 1.
- ✓ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

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[23/10/2026]